# Catchment — Reading List

*A curated, verified reading list — to turn open data into a demand map, a reach table and a defensible plan for where the next dark stores should go.*

Companion to the *Catchment* project · Oct 2026 · every link verified against primary sources · old foundations + current (2023–26) work

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> **How to use this.** You don't need all of it. Read §1–2 for stage 0 (demand), §3 for stage 1 (reach), and §4–5 for stage 2 (the CP-SAT plan and its greedy baseline). §8 is the reference for stage 3 (API and map), §6–7 give the domain and the industry context and the dated store counts for the competitor layer, §9 is a lookup table, and §10 is for the later ops simulator. Operations research, retail geography and q-commerce are the new ground here, so §4 and §6 come first within their stages; you already know CP-SAT mechanics, GIS and MapLibre, so those entries are short.

**Suggested path (≈4 weeks of evenings, alongside the roadmap):** Week 1 — §1 H3 and MAUP, §2 WorldPop, night lights and Overture, while building stage 0 → Week 2 — §3 OSRM, OSM quality in India and the travel-time error papers, for stage 1 → Week 3 — §4 Church & ReVelle, Huff and the competitive-location overview, then §5 the CP-SAT primer and greedy guarantees, for stage 2 → Week 4 — §8 serving stack, §6 for sourcing competitor counts, §7 for one engineering post, then §10 when the first cut ships.

## 1 — H3 & Spatial Indexing (the unit demand is counted in)

**H3: Uber's Hexagonal Hierarchical Spatial Index** — Brodsky, Uber Engineering blog (2018)
*The original statement of why hexagons (one neighbour distance, 12 pentagons, approximate parent-child nesting); read it before you pick the working resolution.*
`uber.com/blog/h3/`

**H3 Documentation: Introduction and Resolution Table** — Uber / h3geo.org
*Cell areas and edge lengths per resolution, plus the traversal and polyfill calls you will use to hex the Gurugram + South Delhi polygon.*
`h3geo.org/docs`

**Rectangular and Hexagonal Grids Used for Observation, Experiment and Simulation in Ecology** — Birch, Oom & Beecham, Ecological Modelling 206(3-4) (2007) paywalled
*The neutral, non-Uber argument for hexagons (uniform adjacency, less edge bias) and where squares still win; it backs your CONTEXT.md choice of "hex".*
`doi.org/10.1016/j.ecolmodel.2007.03.041`

**The Modifiable Areal Unit Problem in Multivariate Statistical Analysis** — Fotheringham & Wong, Environment and Planning A 23(7) (1991) paywalled
*Shows that results change with zone size and shape; the reason to test two H3 resolutions and report how the plan moves, not just pick one.*
`doi.org/10.1068/a231025`

## 2 — Building a Demand Surface (population, affluence, activity; every factor a labelled assumption)

**WorldPop, Open Data for Spatial Demography** — Tatem, Scientific Data 4 (2017)
*What the WorldPop grids are, how they are produced and what their uncertainty is; the base of your demand formula.*
`doi.org/10.1038/sdata.2017.4`

**Disaggregating Census Data for Population Mapping Using Random Forests with Remotely-Sensed and Ancillary Data** — Stevens, Gaughan, Linard & Tatem, PLOS ONE 10(2) (2015)
*The method behind WorldPop's top-down grids; explains why 100 m cells are modelled, not counted, so hex totals inherit that error.*
`doi.org/10.1371/journal.pone.0107042`

**Spatially Disaggregated Population Estimates in the Absence of National Population and Housing Census Data** — Wardrop, Jochem, Bird, Chamberlain et al., PNAS 115(14) (2018)
*The bottom-up counterpart to Stevens; useful for judging how far to trust any single population layer.*
doi.org/10.1073/pnas.1715305115

**Generating Surface Models of Population Using Dasymetric Mapping** — Mennis, The Professional Geographer 55(1) (2003) paywalled
*The classic dasymetric idea: redistribute counts onto the cells where people actually live; you do a version of this when you weight population by buildings.*
doi.org/10.1111/0033-0124.10042

**Annual Time Series of Global VIIRS Nighttime Lights Derived from Monthly Averages: 2012 to 2019** — Elvidge, Zhizhin, Ghosh, Hsu & Taneja, Remote Sensing 13(5) (2021)
*The method paper for the VIIRS product you use as an affluence proxy, with its filtering and its limits.*
doi.org/10.3390/rs13050922

**Which Night Lights Data Should We Use in Economics, and Where?** — Gibson, Olivia, Boe-Gibson & Li, J. Development Economics 149 (2021) paywalled
*Compares VIIRS and the older DMSP lights as a stand-in for economic activity, and says where each breaks down, notably at fine scale and in dense cities.*
doi.org/10.1016/j.jdeveco.2020.102602

**Microestimates of Wealth for All Low- and Middle-Income Countries** — Chi, Fang, Chatterjee & Blumenstock, PNAS 119(3) (2022)
*The Meta Relative Wealth Index method, a satellite-and-ML affluence layer to compare your night-light proxy against (data licence still unchecked).*
doi.org/10.1073/pnas.2113658119

**Overture Maps Places Guide** — Overture Maps Foundation
*Schema, categories and confidence scores for the places layer; read before choosing which categories count as commercial activity and as candidate-site evidence.*
docs.overturemaps.org/guides/places/

## 3 — Road Networks & Travel Time (reach: how long a rider really takes)

**Contraction Hierarchies: Faster and Simpler Hierarchical Routing in Road Networks** — Geisberger, Sanders, Schultes & Delling, WEA 2008 paywalled
*The algorithm OSRM's fast mode is built on; explains why a site-by-hex matrix is cheap to compute after a one-off preprocessing step.*
doi.org/10.1007/978-3-540-68552-4_24

**Real-Time Routing with OpenStreetMap Data** — Luxen & Vetter, ACM SIGSPATIAL (2011) paywalled
*The OSRM paper itself: what it computes, what it ignores (live traffic), and how it is fed by OSM.*
doi.org/10.1145/2093973.2094062

**Route Planning in Transportation Networks** — Bast, Delling, Goldberg, Müller-Hannemann et al., arXiv (2015)
*The survey of speed-up techniques (CH, hub labels, and others); read the sections on one-to-many and matrix queries for your travel-time table.*
arxiv.org/abs/1504.05140

**OSRM Profiles** — Project OSRM (GitHub docs)
*How a profile turns OSM tags into speeds and penalties; this is where a two-wheeler profile and its default speeds live, and what you would edit.*
github.com/Project-OSRM/osrm-backend/blob/master/docs/profiles.md

**OpenStreetMap Data as a Basis for Routing: How Well Does It Really Work?** — Herfort & Würzler, HeiGIT write-up of a FOSSGIS 2026 talk (Apr 2026)
*Reports that max-speed tags are only about 1.9% complete on one Indian corridor, so engines fall back on defaults; a direct warning for free-flow times in India (talk write-up, not peer-reviewed).*
heigit.org/openstreetmap-data-as-a-basis-for-routing-how-well-does-it-really-work/

**Study of Travel Time Variability Using Two-Wheeler Probe Data: An Indian Experience** — Chepuri, Wagh, Arkatkar & Joshi, Proc. ICE Transport 171(4) (2018) paywalled
*GPS-instrumented two-wheeler runs in Surat; one of the few Indian two-wheeler travel-time datasets, a sanity check for your congestion multiplier.*
doi.org/10.1680/jtran.16.00101

**A Comparison of Travel Time Data from Open Source Routing Machine (OSRM) and Uber** — Ghanem & Rabalert, ACM SIGSPATIAL Workshop on Spatial Intelligence for Smart and Connected Communities (2025) paywalled
*A direct OSRM-against-traffic-data comparison; the closest published precedent for the spot-check error you will report (title as in Crossref; ACM page is bot-blocked).*
doi.org/10.1145/3764924.3770898

**Using Desktop GIS for the Investigation of Accessibility by Public Transport: An Isochrone Approach** — O'Sullivan, Morrison & Shearer, Int. J. Geographical Information Science 14(1) (2000) paywalled
*The standard reference for network-time catchments rather than circles, which is your brief's central complaint about spreadsheets.*
doi.org/10.1080/136588100240976

## 4 — Facility Location: The Classics (what a catchment, a cover and a rival mean)

**Optimum Locations of Switching Centers and the Absolute Centers and Medians of a Graph** — Hakimi, Operations Research 12(3) (1964) paywalled
*The origin of the p-median and p-center problems; the baseline vocabulary for "choose N sites on a network".*
doi.org/10.1287/opre.12.3.450

**The Maximal Covering Location Problem** — Church & ReVelle, Papers of the Regional Science Association 32(1) (1974) paywalled
*Your model in its original form: choose p sites to maximise demand within a distance standard; short and readable.*
doi.org/10.1007/BF01942293

**Capacitated Covering Models** — Current & Storbeck, Environment and Planning B 15(2) (1988) paywalled
*Adds store capacity to the covering problem, which is the step from the textbook model to yours.*
doi.org/10.1068/b150153

**Covering Problems in Facility Location: A Review** — Farahani, Asgari, Heidari, Hosseininia et al., Computers & Industrial Engineering 62(1) (2012) paywalled
*A map of covering variants (partial, gradual, backup, capacitated); use it to name what you built and what you left out.*
doi.org/10.1016/j.cie.2011.08.020

**Maximal Coverage Location Problem: Impacts, Significance, and Evolution** — Murray, International Regional Science Review 39(1) (2016) paywalled
*A modern retrospective on MCLP, including its known weaknesses (a hex is either covered or not); good for the honest-limitations section.*
doi.org/10.1177/0160017615600222

**Defining and Estimating a Trading Area** — Huff, Journal of Marketing 28(3) (1964) paywalled
*The Huff model: a customer's chance of choosing a store rises with its attractiveness and falls with distance, divided across all rivals; the cleanest basis for splitting a shared catchment.*
doi.org/10.1177/002224296402800307

**A New Set of Spatial-Interaction Models: The Theory of Competing Destinations** — Fotheringham, Environment and Planning A 15(1) (1983) paywalled
*Fixes a known flaw of gravity and Huff models (clustered stores are not independent); relevant when several of your stores sit close together.*
doi.org/10.1068/a150015

**Static Competitive Facility Location: An Overview of Optimisation Approaches** — Plastria, European Journal of Operational Research 129(3) (2001) paywalled
*Where competitors, market share and cannibalisation enter the location problem, and why competitive models get hard; frames your synthetic competitor layer.*
doi.org/10.1016/s0377-2217(00)00169-7

## 5 — Solving It: CP-SAT & Greedy Guarantees (you know CP-SAT; this is for the covering structure and the baseline)

**CP-SAT Solver** — OR-Tools, Google for Developers
*The official guide; skim the sections on Boolean indicators and objective bounds, then read solver logs, which tell you the optimality gap of each plan.*
developers.google.com/optimization/cp/cp_solver

**The CP-SAT Primer** — Krupke, TU Braunschweig (GitHub)
*Practical modelling and tuning advice, including how CP-SAT compares with MIP solvers and what its parameters do; use it for the 10-second budget.*
github.com/d-krupke/cpsat-primer

**An Analysis of Approximations for Maximizing Submodular Set Functions, I** — Nemhauser, Wolsey & Fisher, Mathematical Programming 14(1) (1978) paywalled
*Proves greedy reaches at least (1 − 1/e), about 63%, of the optimum for max coverage; the theoretical floor your greedy baseline sits on.*
doi.org/10.1007/BF01588971

**The Budgeted Maximum Coverage Problem** — Khuller, Moss & Naor, Information Processing Letters 70(1) (1999) paywalled
*Greedy guarantees when each site has a cost; the nearest result to capacity-limited siting.*
doi.org/10.1016/S0020-0190(99)00031-9

## 6 — Quick Commerce & Dark Stores (the problem domain, from papers and company filings)

**Heuristic for Optimisation of Dark Store Facility Locations for Quick Commerce Businesses** — Kewalramani & Khadilkar, arXiv 2312.11494 (2023)
*The roadmap's second baseline: a heuristic for siting stores under a service radius and a daily order cap; a preprint, so read it critically.*
arxiv.org/abs/2312.11494

**Winning the Race to Customers with Micro-Fulfillment Centers: An Approach for Network Planning in Quick Commerce** — Yang, Ostermeier & Hübner, Central European J. Operations Research 32(2) (2024) paywalled
*A peer-reviewed network-planning model for q-commerce; compare its objective and constraints with yours.*
doi.org/10.1007/s10100-023-00893-x

**Mitigating Disruption Impact in Q-Commerce Through Optimization of Dark Store Resilient Portfolio** — Raj, Das & Sawik, Transportation Research Part E 205 (2026) paywalled
*Dark-store selection with disruption and backup stores; relevant to your congestion lever.*
doi.org/10.1016/j.tre.2025.104518

**Dark Stores in the City of Light: Geographical and Transportation Impacts of 'Quick Commerce' in Paris** — Buldeo Rai, Mariquivoi, Schorung & Dablanc, Research in Transportation Economics 100 (2023) paywalled
*An empirical look at where dark stores open and how they cluster in a real city, without operator data.*
doi.org/10.1016/j.retrec.2023.101333

**Swiggy Annual Report FY 2024-25** — Swiggy Limited (2025)
*Primary source for Instamart store counts, store area and contribution margin; every competitor count you use should cite a document like this, with its date.*
swiggy.com/corporate/wp-content/uploads/2025/07/Swiggy-Annual-Report-FY-2024-25.pdf

**Swiggy Quarterly Shareholder Letters (e.g. Q4 FY2026)** — Swiggy Limited (2024-26)
*Quarter-by-quarter store counts and unit economics, for the dated figures a synthetic competitor layer needs.*
swiggy.com/corporate/wp-content/uploads/2026/05/Q4-FY2026-Shareholder-letter.pdf

**Eternal: Shareholders' Letter and Results** — Eternal Limited (Blinkit and Zomato)
*The index of Eternal's quarterly letters and results, where Blinkit store counts and margins are reported.*
eternal.com/investor-relations/results/

## 7 — How Industry Builds This (engineering posts from real companies)

**Next-Generation Optimization for Dasher Dispatch at DoorDash** — Jin, DoorDash Engineering (Feb 2020)
*Moves dispatch from one-to-one matching to a mixed-integer program that can build multi-order routes; the origin story for your later rider-batching stage.*
doordash.engineering/2020/02/28/next-generation-optimization-for-dasher-dispatch-at-doordash/

**Using ML and Optimization to Solve DoorDash's Dispatch Problem** — Weinstein, DoorDash Engineering (Aug 2021)
*ML predicts how an order will unfold, then an optimisation layer assigns; shows how predicted times feed a solver, as your travel-time table feeds CP-SAT.*
doordash.engineering/2021/08/17/using-ml-and-optimization-to-solve-doordashs-dispatch-problem/

**The Swiggy Delivery Challenge (Part One)** — Gupta, Swiggy Bytes (2018)
*Swiggy's own breakdown of delivery time into assignment delay, first mile and last mile, in an Indian operating context.*
bytes.swiggy.com/the-swiggy-delivery-challenge-part-one-6a2abb4f82f6

**Pharos: Searching Nearby Drivers on Road Network at Scale** — Grab Engineering (2020)
*Nearest-by-road-time search rather than straight-line distance, at production scale; the engineering counterpart of your OSRM reach table.*
engineering.grab.com/pharos-searching-nearby-drivers-on-road-network-at-scale

## 8 — The Serving Stack (API, map, hex layers)

**FastAPI Documentation** — Ramírez
*Request models and start-up loading for POST /plan; the lifespan and background-task sections matter for holding the travel-time table in memory.*
fastapi.tiangolo.com

**MapLibre GL JS Documentation** — MapLibre
*You know it from the rooftop project; here you need data-driven fill styling for hex choropleths and layer toggling for two scenarios side by side.*
maplibre.org/maplibre-gl-js/docs/

**PMTiles Concepts** — Protomaps
*Single-file static vector tiles you already know; relevant if the demand and catchment layers are too big to ship as GeoJSON.*
docs.protomaps.com/pmtiles/

**deck.gl H3HexagonLayer** — OpenJS Foundation / vis.gl
*Draws hexes straight from H3 indices, with no polygon geometry to ship; the lightest way to show demand per hex with MapLibre as the base map.*
deck.gl/docs/api-reference/geo-layers/h3-hexagon-layer

**h3-js** — Uber (GitHub)
*H3 in the browser, for cell-to-boundary and hover lookups if you draw hexes with MapLibre alone.*
github.com/uber/h3-js

**PostGIS Documentation** — PostGIS project
*Only if the area grows to all NCR and the travel-time table needs spatial queries rather than in-memory arrays; not needed for the first cut.*
postgis.net/documentation/

## 9 — Datasets & Tools Lookup (a table you will return to)

**WorldPop** — University of Southampton, CC BY 4.0
*≈100 m population grids; the base demand layer.*
worldpop.org

**VIIRS Nighttime Lights** — Earth Observation Group, Colorado School of Mines
*Annual and monthly composites for the affluence proxy; check the product page for terms and the version you use.*
eogdata.mines.edu/products/vnl/

**Overture Maps Documentation** — Overture Maps Foundation
*Places, buildings and download tooling; per data-sources.md, Places carry CDLA-Permissive-2.0.*
docs.overturemaps.org

**Geofabrik: India Extract** — Geofabrik GmbH (OSM, ODbL)
*The daily India OSM extract that you clip to the area before building the OSRM graph.*
download.geofabrik.de/asia/india.html

**Continental-Scale Building Detection from High Resolution Satellite Imagery (Open Buildings)** — Sirko, Kashubin, Ritter, Annkah et al., arXiv 2107.12283 (2021)
*The paper for Google's Open Buildings dataset (CC BY 4.0), a built-up-density alternative to Overture buildings.*
arxiv.org/abs/2107.12283

**Project OSRM** — OSRM project
*Routing engine docs: building the graph, running osrm-routed and the table service for many-to-many times.*
project-osrm.org

**Osmium Tool** — Jochen Topf / OSM community
*Clipping the India extract to your area with a polygon before OSRM, as the roadmap's 8 GB machine requires.*
osmcode.org/osmium-tool/

**h3-py Documentation** — Uber
*The Python H3 bindings (v4 API) used for the demand table and hex lookups.*
uber.github.io/h3-py/intro.html

## 10 — Later: Ops Simulation & Routing (rider batching, VRP and dispatch, for after the first cut)

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**Algorithms for the Vehicle Routing and Scheduling Problems with Time Window Constraints** — Solomon, Operations Research 35(2) (1987) paywalled
*Origin of the Solomon VRPTW instances; delivery promises are time windows, so this is your benchmark language.*
doi.org/10.1287/opre.35.2.254

**Solomon Benchmark (VRPTW)** — SINTEF
*The instance files and best-known solutions, to test any routing code against before trusting it on your own data (page is bot-blocked).*
sintef.no/projectweb/top/vrptw/solomon-benchmark/

**Hybrid Genetic Search for the CVRP: Open-Source Implementation and SWAP* Neighborhood** — Vidal, Computers & Operations Research 140 (2022) paywalled
*The state-of-the-art heuristic that PyVRP builds on; read it to know what a good rider-routing solver looks like.*
doi.org/10.1016/j.cor.2021.105643

**PyVRP: A High-Performance VRP Solver Package** — Wouda, Lan & Kool, INFORMS J. Computing 36(4) (2024)
*A pip-installable VRP solver with time windows; the likeliest drop-in for rider batching in the simulator.*
doi.org/10.1287/ijoc.2023.0055

**OR-Tools Routing Library** — Google for Developers
*Routing with time windows and capacities in the tool you already use; a second option next to PyVRP.*
developers.google.com/optimization/routing

**The Meal Delivery Routing Problem** — Reyes, Erera, Savelsbergh, Sahasrabudhe & O'Neil, Optimization Online (2018)
*Defines the dynamic courier-assignment problem on real restaurant-delivery data; the closest open formulation to instant-delivery dispatch (preprint; journal version not confirmed).*
optimization-online.org/2018/04/6571/

**The Restaurant Meal Delivery Problem: Dynamic Pickup and Delivery with Deadlines and Random Ready Times** — Ulmer, Thomas, Campbell & Woyak, Transportation Science 55(1) (2021) paywalled
*Handles uncertainty in preparation times; your picking stage is the same random ready time.*
doi.org/10.1287/trsc.2020.1000

**Meituan's Real-Time Intelligent Dispatching Algorithms Build the World's Largest Minute-Level Delivery Network** — Liang, Luo, Duan, Li, Liao et al., INFORMS J. Applied Analytics 54(1) (2024) paywalled
*A practitioner account of minute-level dispatch at national scale, combining OR and ML; shows what the real system optimises.*
doi.org/10.1287/inte.2023.0084

**SimPy Documentation** — SimPy team
*Process-based discrete-event simulation in Python; the natural engine for orders, picking and riders as event processes.*
simpy.readthedocs.io

## What's Still Unverified

**Company figures:** store counts and margins in the Swiggy and Eternal documents were not read in full or checked against each other; reported counts vary by source and date, so record each number with its source and date, as data-sources.md already requires. Individual Eternal letters were not opened; only the results page was confirmed.

**arXiv 2312.11494:** found only as an arXiv preprint (a Research Square copy also exists); no peer-reviewed version was found.

**Reyes et al. (2018):** confirmed as an Optimization Online preprint; a journal version was not confirmed. A Transportation Science paper on the same problem is by different authors (Yildiz & Savelsbergh, 2019) and is not listed.

**Ghanem & Rabalert (2025):** bibliographic data confirmed via Crossref; the study area and findings were not read, so check them before citing the result.

**India two-wheeler speeds:** no confirmed city-level two-wheeler speed survey for Delhi NCR was found; the Surat study and the OSRM-versus-traffic comparison are the nearest sources, so the congestion multiplier remains an assumption to calibrate by your spot-check.

**Meta Relative Wealth Index:** the method paper is listed, but the dataset licence has not been checked.

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**Every link verified to resolve against a primary source (Oct 2026).** DOI metadata (authors, year, venue) was checked against Crossref and arXiv records; web pages were fetched or confirmed by search. Bot blocks prevented direct fetches of the Uber blog (HTTP 406), the SINTEF Solomon pages (403), the DoorDash engineering posts (the doordash.engineering URLs redirect to a host that returns 403), the ACM Digital Library (403), the Swiggy Bytes host (connection timed out), and several publisher sites behind DOIs (Sage, T&F, MDPI, INFORMS, PNAS and ACM return 403 to scripts; the DOIs themselves resolve and their metadata matches Crossref); the Uber, SINTEF, DoorDash and Swiggy Bytes posts were confirmed through search results, and the ACM paper through Crossref, and are kept. Yellow tags mark paywalled items. Corrected on the way: two DOIs first guessed wrongly (the Farahani covering review and the O'Sullivan isochrone paper) were replaced by the Crossref-confirmed ones. Dropped as unconfirmed or unusable: engineering posts from Zepto and Instacart (no primary post found), Swiggy Bytes ETA posts (no URL confirmed), the CP-SAT-LP paper (no DOI found), the Eternal annual report PDF (not located), a Sydney Google-versus-Uber travel-time comparison (not about routing engines), and all SEO and consultancy blogs on q-commerce. Companion document: *Location Intelligence & Logistics — Foundations*.